

[G0R72A] Data Visualisation Project Report

October 29, 2024

Part 1 Metadata

- Version: implementation (17/5/2024)
- Students: Salma Loukili r0853837 KU Leuven – Tom Segers r0918109 KU Leuven
- Group number: group 16
- Dataset: d.e.a.d.

Part 2 Project Description

This report explores the D.E.A.D. dataset, which includes data on customer profiles, product transactions, and sales performance across different regions within a retail environment. We aim to use data visualization techniques to understand customer behavior, product trends, and regional sales strategies. By analyzing this dataset, we hope to uncover patterns and insights that can inform better business decisions.

D.E.A.D. is an adventure equipment company and the dataset delves into the following tables:

- Customers: This table contains detailed attributes like Account Code and Account Manager, crucial for understanding the link between sales performance and customer service efficiency. The differentiation between private individuals and professional buyers, as well as the presence of a 'key account' status for larger professional clients, provides a unique lens through which purchasing behaviors and client service needs can be analyzed.
- Products: With each product categorized by Type and Subtype and distributed across three distinct Business Units, the dataset offers a structured approach to assess product performance.
- Regions: This part of the dataset details the Territories within the Forgotten Realms, enabling an analysis of market penetration and regional sales effectiveness.
- Orders: Including comprehensive details like Cart Price and Product Price, alongside quantities purchased, this dataset section tracks the total revenue generated per transaction and the unit price of products sold. It's instrumental in identifying top-selling products, understanding average order values, and analyzing sales trends over time.

By linking customer data (Region, Private, Key Account or no Key Account) with specific product details (Type, Subtype, Business Units) and overlaying this information with sales data from the Orders section (Cart Price, Product Price), we can unveil patterns in purchasing habits. Moreover, analyzing these patterns against the backdrop of geographic territories allows for a comprehensive view of D.E.A.D.'s operational landscape. To guide us through this project, a selection of topics has been created with questions that we want to focus on answering and providing more insight into. Addressing these questions through data visualization will illuminate critical insights into D.E.A.D.'s operational and strategic performance, facilitating data-driven decisions to enhance market reach, optimize product portfolio, and ensure the highest levels of customer satisfaction in the adventurous domain.

1. Which product types and subtypes are most favored across different territories?
2. How is the pricing of different products affected by the region and quantity purchased?
3. How is the sales performance spread over the years?

Part 3 Visual Design

In the initial phase of our visualization project, my group and I engaged in an extensive brainstorming session to explore various ideas and approaches for representing the dataset visually. Through collaborative discussion and ideation, we generated a diverse range of concepts and potential visualization designs. These sketches were inspired by our collective understanding of the dataset's key attributes, such as orders, product types, sizes, delivery times, and regions.

Each sketch aimed to capture different aspects of the data while considering factors like clarity, interpretability, and visual appeal. After careful deliberation, we narrowed down our pool of sketches to around 20 promising candidates, which exhibited the most potential for effectively conveying meaningful insights from the dataset.

These sketches served as the foundation for further refinement and iteration as we moved towards the final stages of our visualization development process. The main focus was to provide interactive designs that give us a deeper insight into the data while making it a fun experience for the user.

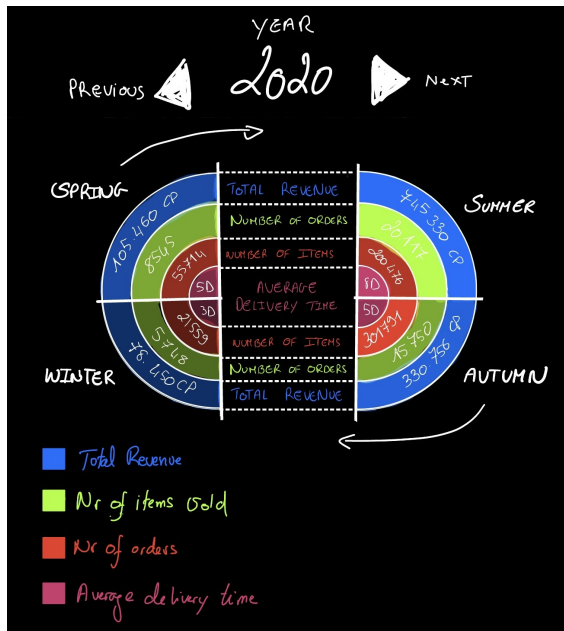


Figure 1: Sales and delivery times per season

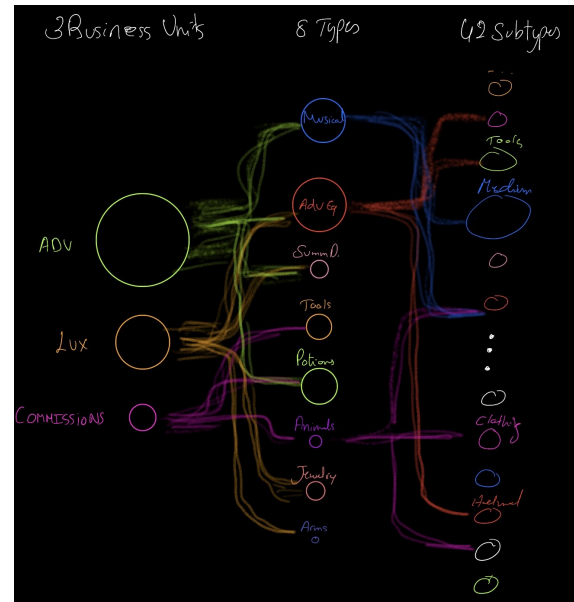


Figure 2: Sankey diagram of the products and sub-products

Our first selected candidate, in figure 1 is a multidimensional visualization that leverages color to represent different dimensions within the dataset. Each color corresponds to a distinct dimension, while the brightness of the color signifies the quantity associated with that dimension, this is in combination with the actual simplified numbers that are displayed simultaneously. The idea is to provide an overview of the sales distinguished by the four seasons of the year. The design allows the user to also interactively display the data for different years.

Our second chosen sketch is a Sankey diagram found in figure 2 that portrays the categorical dimension of the products within the dataset through a network visualization. In this visualization, each category is represented by a distinct node, with lines connecting nodes to illustrate the relationships between categories. The node size represents the distributions of the categories. This diagram is a simple and intuitive way of displaying the connections between the categories and understand the products.

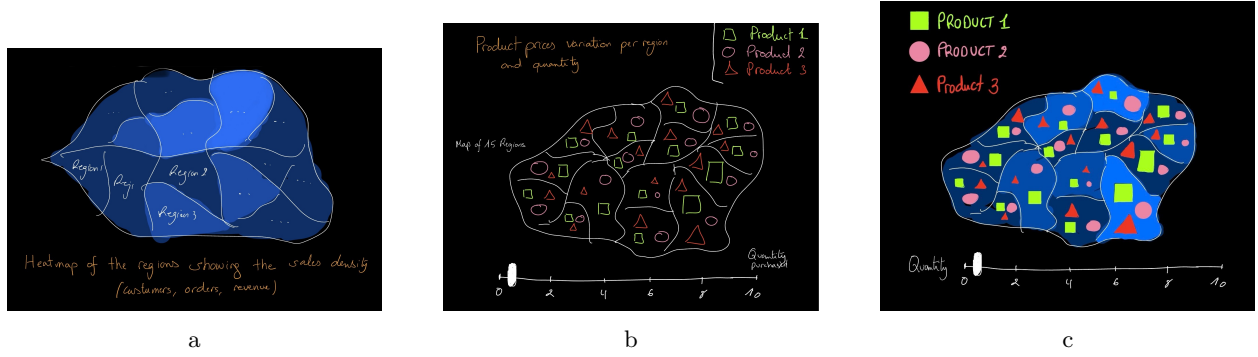


Figure 3: Prices of products with varying region and quantity purchased

Our 3rd selected sketch in figure 3 adopts a straightforward yet visually impactful approach to representing the dataset, focusing on regions and product prices. In this map visualization, which the user can directly interact with, showcases the variation of the price of a certain product under circumstances like region and quantity purchased. The drag bar at the bottom, allows the user to see the data from the extra dimension of the quantity purchased of said product. That way, we can easily visualize how the quantity and the region can impact the price of a product.

The creation of this visualization can perfectly illustrate the principle of diverge-emerge-converge. The colors of the regions are of different intensities, which come from a heatmap sketch that portrays the density in sales. This heatmap was then combined with the map of regions which contains the shapes (products) in different sizes (prices). The interactive drag bar was also added at the end for an extra dimension (quantity).

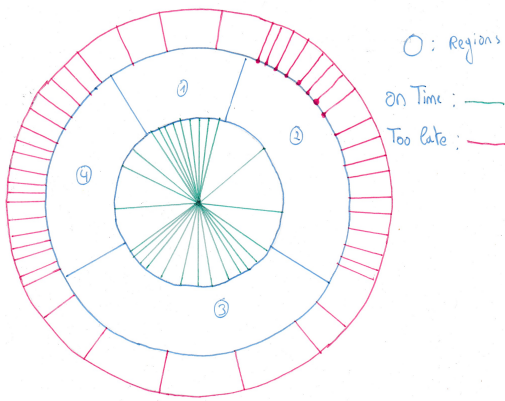


Figure 4: Delivery times per region

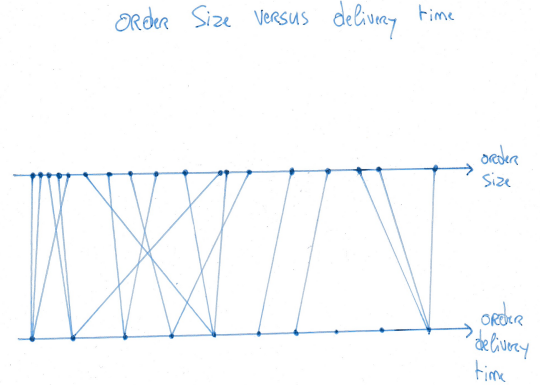


Figure 5: Delivery time vs order size

The 4th sketch shown in figure 4 dives into the relationship between regions and delivery time. It's all about figuring out if there are any regions where delivery needs a boost. We think there's more to explore here. Maybe adding some extra dimensions could help uncover what's causing the delivery delays. This sketch has got potential to be even more insightful with a bit of expansion and digging into other factors.

The 5th sketch found in figure 5 delves into the dynamics of delivery time, this time through the lens of order size. It represents an additional layer of analysis aimed at uncovering the intricate relationship between these variables. However, we're considering enhancing this visualization by introducing a color-coded dimension. By doing so, we aim to discern the specific product types that might contribute to delivery delays. This proposed refinement seeks to augment the depth and precision of our visual representation. With the incorporation of colour, this sketch possesses the potential to offer profound insights into underlying patterns and correlations within the dataset.

The 6th design in figure 6 resembles a network of connected stars in the night sky. Each dot represents

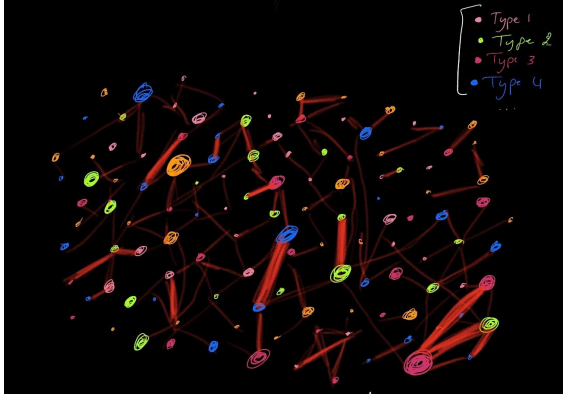


Figure 6: Which products are bought together

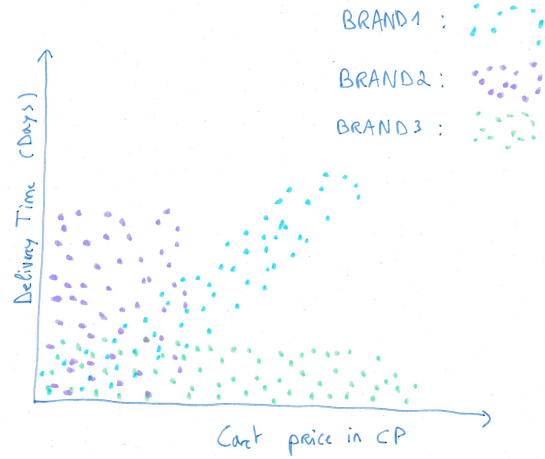


Figure 7: Delivery time in function of cart price and brand name

a single product, and its size represents its sales share. Products of the same type have the same color. The diagram showcases which products are often bought together, by connecting the dots with lines for each time two products are purchased together in the same order. The thickness of these lines represent thus the frequency of two products being bought together. The idea is to make this visualization also user interactive, by hovering over any dot would show the complete name and type of a product.

Sketch 7 highlighted by figure 7 offers a graphical depiction of delivery time relative to cart prices, with each dot representing an individual order. Notably, the colors assigned to each order denote its originating brand, facilitating the observation of potential clustering behavior. Through this visualization, our aim is to discern any patterns or trends in delivery time, particularly with respect to varying cart prices and brand origins. By examining potential clusters, we hope to identify instances where certain brands may struggle to meet delivery expectations. This approach enables us to gain deeper insights into the performance of different brands within the dataset and their respective impacts on delivery efficiency.

Final Decisions

For the last step of this section, we have decided to implement sketches that deem the most insightful and highlight the most the diverge-converge methodology in our design process. Firstly, Sketch 3 converges from two different design ideas and provides the most engaging visualization and great insights into the pricing strategies of different products.

The second sketch we've selected to work with is Sketch 2, found in figure 2. This choice stems from its ability to provide a good understanding of the links between dimensions, offering initial insights into potential relationships. While Sketch 2 already offers a solid overview of the dataset, we aim to enhance it by incorporating elements from other sketches. Specifically, we'll add size to each circle to represent the number of orders containing the product type. Additionally, we'll introduce color to each circle, with the brightness indicating the mean delivery time of the product type.

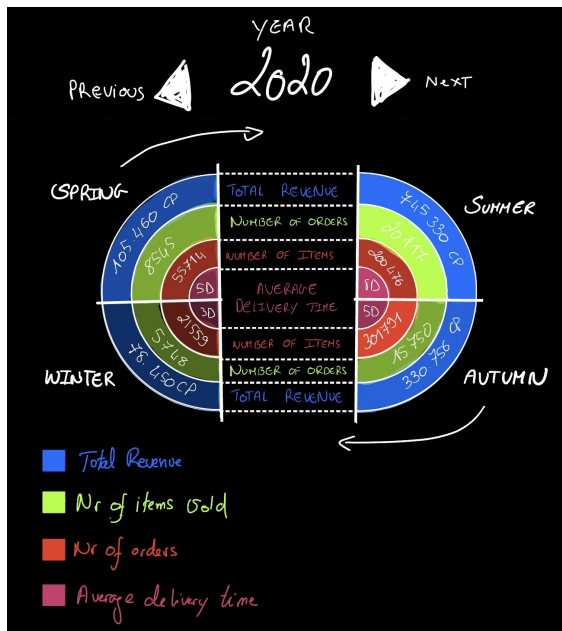
Lastly, sketch 1, in figure 1 is a basic yet super impactful visualization that we think is going to provide great insights on the sales performance over all the years. Traveling and adventure taking tend to have seasonal trends, so we opted for a design that shows an overview of sales for each season of each year in the dataset. This will help identify trends and purchasing habits over the years.

Part 4 Implementation

Following the design phase, two sketches have been chosen to be implemented using the Svelte framework. This incorporates the use of HTML, CSS, d3 and JavaScript to create the responsive visualizations. The first sketch that was chosen is sketch 1, in which a total overview of the sales, revenues, orders and delivery times are summed up in a circular design for each year for which there is data on the database. The circular shape was chosen as the design focuses on highlighting these attributes per season. This provides not only information of the total activity per year, but also which seasons these activities are most prevalent. This is relevant for our database, since it is easy to relate traveling and adventure equipment to certain seasons. The design aims to reveal important insights into the trends of orders and their volumes depending on the season.

The second sketch to be implemented is sketch 3. This design is particularly more complex and provides an engaging interaction aiming to reveal the trends behind the pricing of the products and how they are affected by the regions and quantity purchased. The main idea behind this sketch is to display a certain number of products on the regions map, while representing their prices by a certain shape, this shape has a size proportional to the price and it changes by region and by quantity purchased. The interaction aspect lays within the slide bar, by which the user gets to change the quantity of said product. The different regions are also colored with different intensities, representing the volume of sales in that region. This design was chosen as it provides both an engaging as well as an insightful visualization of the pricing data of the products. Due to many products being available, an extra feature can be added to give the user an extra dimension of freedom by implementing a drop down menu aiding to select the products they wish to display on the map, providing both a comparison aspect as well as more interactivity.

Implementation 1



Intended design: Sketch 1



Actual visualization

Figure 8: Implementation 1

The intended design can be found in the figure left in 8. The handmade sketch consists of four quarter circles representing each a season and buttons to navigate through the years. The legend is included for more clarity. The intended design has a responsive effect, in which the color intensity of the circle bands changes according to the labeled numbers on them. Ideas to provide a yearly overview, besides the seasonal

split was also considered.

The actual implemented design can be found on the right in 8. The visualization is almost identical, with some color changes and button enhancements. The interaction is the same, the user can display a yearly seasonal overview of the sales of each year, by using the buttons to navigate through the years. However, the responsive effect, displayed by the color intensity change of the circle bands, is not present in the actual design. The data turned out to be very uniformly distributed, with very few variation, this made the color intensity change almost invisible. Finding a way to make this feature noticeable was quite challenging.

The implementation of this chart starts in Python, the data must be processed into an appropriate format for the design. The code processes and analyzes order data using Python's pandas library. It starts by loading and merging two CSV files, *orders.csv* and *regions.csv*, based on the *Territory* column. *OrderDate* and *DeliveryDate* are converted to datetime objects, and *Quantities* are normalized by splitting, converting to integers, and summing them. Delivery time in days is calculated, and *Year* and *Season* columns are extracted from *OrderDate*, categorizing months into seasons. The data is grouped by *Year* and *Season*, and key metrics—total revenue, total orders, total products, and average delivery time are aggregated. Finally, the summary data is converted to JSON format and saved to *summary_data.json*. The code effectively consolidates, processes, and summarizes order data by year and season.

The JSON data was then queried from the final implementation, which constructs an interactive data visualization webpage using HTML, CSS, and JavaScript. It presents sales metrics by season across multiple years through SVG graphics. The HTML defines the structure with headings, navigation buttons, and SVG elements, which include circles, text paths, rectangles, and lines for visual representation. CSS provides styling for a black background, centered layout, and specific colors for different metrics.

JavaScript stores sales data in an object and updates the visualization dynamically. Event listeners on "Previous" and "Next" buttons allow users to navigate between years, triggering the *updateChartData* function to refresh the displayed data. The SVG elements are manipulated to reflect the current year's data, with circles representing different metrics (total revenue, total orders, total items sold, average delivery time) and text paths annotating these circles.

Implementation 2

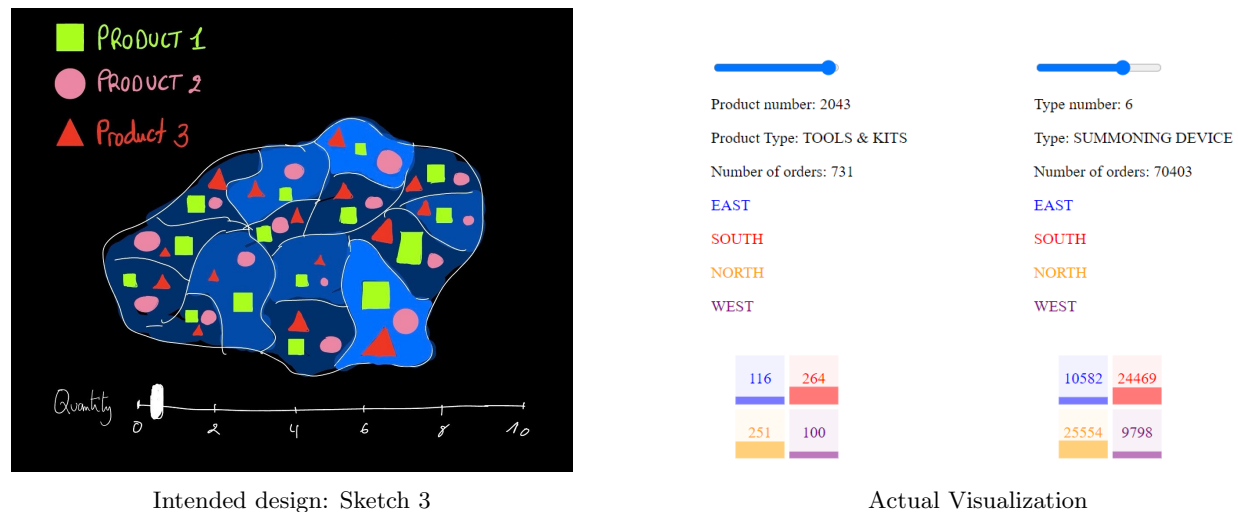


Figure 9: Implementation 2

The actual implementation might appear significantly different from the sketch, but the underlying concept remains the same. While the sketch displays multiple products simultaneously, I opted for a more interactive design where users can manipulate bars representing individual products and product types. Additionally, alongside the implementation for all products, I provided an alternative visualization allowing users to select a product type, offering a broader overview of the various product categories.

As previously explained, the interaction available allows users to adjust two different bars corresponding to distinct products and types. Any change of the bar gets picked up very fast because of the dynamic code '\$:' provided by svelte. This is a very nice tool that reacts faster than you would expect. Important to say about this is that I did the computations in matlab and not in svelte, each value of the bar represents a product (respectively product type) which requires a run through the entire order dataset (which contains 280000 rows). As you can imagine, this process takes some time and temporarily freezes the screen in Svelte. To address this issue, I precomputed all the necessary values for every product (or product type) in MATLAB and stored them in separate .csv files. Now, rather than Svelte recalculating everything on the fly, it simply accesses the precomputed values from these .csv files, resulting in smoother performance without any screen freezes.

Aside from visual aesthetics, the visualization effectively communicates the intended information. However, if we had the assistance of a skilled programmer, we could consider incorporating additional visualizations, such as delivery time data. While the number of orders provides insight into crucial areas, understanding regions with poor delivery times could also be valuable.

All code can be found in the github repository: <https://github.com/salmaloukili/Data-Visualization.git>

Both visualizations are demonstrated in this Youtube video: <https://youtu.be/ymCGjq0srE4>

Part 5 Findings

For the DEAD dataset there are only 2 things that were different from our expectations. First of all, upon inspecting the regions dataset, the area column had some missing values and some values different from North, South, East or West. Secondly, in the orders dataset, there were products in some orders that are not in the product data set.

The findings from the second implementation are that the number of orders from the South and North are much higher than the other two areas. This suggests that these areas are bigger or have more clients. Of course, for the company it is interesting to spend more time and money on these areas first rather than spending all resources uniformly.

Additionally, the dataset seems to be very bland and uniform, it was very challenging to find interesting trends that could be visualized nicely.

Part 6 Reflections

Salma: This project has been a really nice learning experience, my favourite part was sketching the designs and combining different ideas. It is fascinating how far our imagination can take us when we let go of the traditional ideas of bars and pie charts and scatter plots. The thing I am most proud of is learning how to think outside of the box of ordinary data visualization plots and tools. I am also very proud of the skills I have gained during the implementation of my design. While having had multiple experiences with HTML, CSS and JavaScript, coding the circular chart was nothing but challenging. I am proud of implementing it and using the SVG library to do so. It really felt like I was creating something from the bare scratch of nothing. The thing I am the least proud of in this project is not using Svelte for the implementation part. It seemed quite of a challenging new framework to use and although I was always resorting to the tutorial compiled for us by the prof, I wasn't always successful at understanding this framework. Leaving the implementation to the last minute didn't help with this and due to time constraints, I had to resort to just coding the visualization in a basic HTML file. The dataset wasn't the best thing about this project either, as it was extremely hard to interpret and work with sometimes. It would be better to use less fake data for such a project, where we can actually identify and analyze trends.

Part 7 Individual Contributions

This project was created by a group of two students, Tom Segers and Salma Loukili. Although there was a significant conflict of schedules and campuses, the collaboration was smooth and efficient. Most meetings for the implementation part were held online, while we deemed it necessary to meet in person for the design phase. The process of exploring the data, collecting ideas and merging them into sketches, using the diverge-converge methodology, was a unique experience that is hard to replicate via an online encounter.

The sketches with a white background were created by Tom while those with a black background were created by Salma. However, the path taken to reach these designs was done collaboratively. For the implementation, the split was even, we both took on an individual journey to make a visualization each, Salma created the first implementation in figure 8 while Tom worked on the second implementation in figure 9. While keeping a close update to each other about the state of the design and implementation, the report was written on a shared document, where each of us wrote the part that corresponds to their own sketch or implementation.

Appendix A: Sketches Overview

